The Compute Floor of Data Mixing: When Loss-Based Reweighters Start to Drain Easy Synthetic Text

Shayan Salehi
University of Waterloo
Waterloo, Ontario, Canada
s4salehi@uwaterloo.ca

Iman Mohammadi
University of California, Irvine
Irvine, California, United States
imohamm1@uci.edu

Abstract

Automatic data-mixing methods are usually evaluated with pretraining sweeps from 70M to billions of parameters, which leaves a basic question open: how much compute does it take to tell these methods apart? We study this on a pool split by provenance, 9 real Pile subsets and 9 synthetic instruction corpora, where every method receives the same per-domain validation-loss signal and we record how it shifts the synthetic share of the mixture. At 160M parameters (2.30×10^{17} FLOPs per run), the adaptive methods separate into two families across three seeds: loss-monotone reweighters move mass toward the harder real text (Group-DRO to a synthetic share of 0.20, DoReMi-style to 0.37), while interaction and gradient estimators stay near the 0.50 start. The split follows a training-induced loss gap that makes real text the high-loss provenance. Two cheaper tiers locate a compute floor: the split is already clear at 1.2×10^{16} FLOPs, but at 2.5×10^{14} FLOPs it becomes seed-dominated, with Group-DRO showing no net direction and DoReMi-style reversing. Held-out test loss gives the least-moving method, Aioli, the lowest mean, a result we hedge given $n = 3$ and the confound of online training. The protocol diagnoses which mixture a policy selects, not final model quality; all code and artifacts are released at github.com/mixing-floor/compute-floor.

1 Introduction

Picking the mix of pretraining sources used to be hand-tuning; now an algorithm picks it. DoReMi (Xie et al., 2023a), RegMix (Liu et al., 2025), and Aioli (Chen et al., 2025) each read domain weights off a learned signal, and a survey literature has piled up around mixture choice in the data pipeline (Albalak et al., 2024). Validation means sweeping proxy or full pretraining: RegMix regresses over a crowd of small proxy runs, DoReMi pairs a reference with a Group-DRO student, Aioli sweeps probes. Reported scales run 70M up into the billions. And because scaling stays smooth, roughly predictable (Kaplan et al., 2020), the hope is you read mixing behavior off the big runs. So how these methods behave, and where they split, mostly gets studied by groups who can foot the sweeps (extended related work in Appendix E).

A small-scale venue suits a narrower question: which separating property surfaces and reproduces cheaply? We track the direction a method nudges the share of a flagged data category once every candidate sees identical signal on a shared pool. Provenance does the splitting. 9 real pretraining subsets from the Pile (Gao et al., 2021) sit beside 9 synthetic instruction corpora, each synthetic example wrapped in a fixed instruction template. We picked that heterogeneity deliberately, since it hands the methods something concrete to disagree over.

Fold synthetic instruction data into a pretraining mix and some automatic policies will starve it to zero. What decides this isn't the optimizer but the trajectory: which provenance the online-trained model starts finding harder as it goes. That signal is already legible at 160M parameters. The reproducibility point follows. A difference the literature pins down

only by sweeping from 70M to 7B shows up seed-stable on a budget anyone can run, so frontier scale buys nothing here. Read the protocol as a record of which mixture each policy picks, not a quality ranking. The null that no method consistently beats stratified sampling (Chen et al., 2025) holds in our runs (Section 5), and the directional reading survives it.

The protocol is compute-light and reproducible: run the candidate methods on a provenance-heterogeneous pool, freeze the signal (per-domain validation loss), and watch how each one moves the synthetic share. At Pythia-160m (Biderman et al., 2023) and roughly 2.30×10^{17} FLOPs per run, six methods fall into two families across three seeds (Figure 1a). The body carries full per-run FLOP accounting (Table 1) across the three tiers we run, which pins a compute floor: the split holds at the Intermediate tier (10^{16} FLOPs on pythia-70m) yet washes into seed noise at the cheapest tier (Section 5). The released artifacts reproduce at the listed cost. As for what we find, loss-monotone reweighters push the synthetic share down (Group-DRO to 0.20, DoReMi-style to 0.37), while the interaction and gradient estimators barely budge from uniform. A held-out test-loss read (hedged at $n = 3$, no static retrain) hands the least-moving method (Aioli) the lowest mean, beating only DoGE and Skill-It (Figure 1c).

2 Setup and protocol

Every method runs the identical protocol, so the only variable across runs is the method itself.

Pool. We assemble 9 real and 9 synthetic domains (listed in Appendix Table 8), each feeding fixed-length token blocks. A fixed instruction template wraps the synthetic examples (`### Instruction: / ### Response:`); real text stays bare. "Synthetic" names this instruction-formatted, assistant-style class: most of its corpora are model-generated, but the slot labeled Self-Instruct holds the human-written Dolly-15k (Conover et al., 2023), ShareGPT pairs human prompts with model responses, and Open-Platypus draws largely on human-authored problems. The scenario we call S2 holds 9 real and 9 synthetic, so every method opens at a synthetic share of exactly 0.50; that setup alone pries the split apart.

Shared signal and read-off. Mixing is sampling. Each step draws a batch by multinomial(batch size, p) over domains. Every online method moves p off one signal, per-domain validation loss, so we compare the update rule and not who got luckier with the input. We log p each round and report, per method, its synthetic-share trajectory: the mass $\sum_{i \in \text{synth}} p_i$ on the 9 synthetic domains (with $\sum_i p_i = 1$ over all 18), and the L_2 distance of the final p from uniform.

Methods and baselines. Six policies, ordered by what their update signal reads. Aioli (Chen et al., 2025) builds an interaction matrix from probe sweeps, then runs exponentiated-gradient descent on its column sums. DoGE (Fan et al., 2024) reads gradient alignment. Skill-It (Chen et al., 2023) does both, pushing a learned skills graph against current loss. Group-DRO (Sagawa et al., 2020) and a DoReMi-style variant (Xie et al., 2023a) move loss-monotonically; the DoReMi-style membership hinges on the frozen reference (Appendix A). Stratified ($p = 1/k$) just stays put, our control. Update rules, per-method notes, and the Aioli code (the Full tier runs a stateful EMA path, not the reference's stateless reset) sit in Appendices A and F. No policy feeds a headline number.

Models and budget. We use HuggingFace Pythia pretrained checkpoints (Biderman et al., 2023) rather than training from scratch: pythia-160m for the Full tier, pythia-70m for the two lighter tiers. We approximate FLOPs per run as $6ND$, where N is total parameters and $D = \text{max_steps} \cdot \text{batch} \cdot \text{ctx}$ tokens. Counting total, not the ∼85M non-embedding parameters, overshoots, so read each budget as a ceiling. Table 1 lists the three tiers, each over three seeds. Every headline number comes from the Full tier; the pythia-70m tiers locate where on the budget axis the split first appears (Section 5). Every run fits on one GPU: ours used a single NVIDIA A100 (80 GB), at a few GPU-hours per seed for all six methods, and the Minimal tier finishes in minutes. The point: the directional split holds across three seeds at a single-GPU budget. Fixing a small budget is fine when the question is whether the phenomenon appears at all, not how it extrapolates, and behavior at one scale need not

Table 1: Tiered configs and per-run FLOP budget. FLOPs/run $\approx 6ND$, $D = \text{max_steps} \cdot \text{batch} \cdot \text{ctx}$. **All three tiers were run** over three seeds; the Full tier (pythia-160m) supplies the reported numbers, and the two pythia-70m tiers map the scale-to-signal curve (Section 5). The last two columns give the two loss-monotone methods' final synthetic share (Group-DRO, DoReMi-style; start 0.50): the split is gone at the Minimal tier (Group-DRO seed std 0.35, DoReMi-style reversed) and present from the Intermediate tier on, while the interaction and gradient methods stay near 0.50 at every tier. DoGE and Skill-It carry one-time estimation surcharges over the Full-tier base, raising their per-run totals to 2.37×10^{17} (+3%) and 2.87×10^{17} (+25%); the other four sit at the 2.30×10^{17} base, all far under the 10^{20} per-run cap (Appendix F).

Config	Model	ctx	batch	max_steps	D (tok/run)	FLOPs/run	GDRO	DoReMi
Minimal	pythia-70m	2048	2	144	5.9×10^5	2.5×10^{14}	0.50	0.75
Intermediate	pythia-70m	1024	4	7200	2.95×10^7	1.2×10^{16}	0.24	0.37
Full (reported)	pythia-160m	2048	8	14400	2.36×10^8	2.30×10^{17}	0.20	0.37

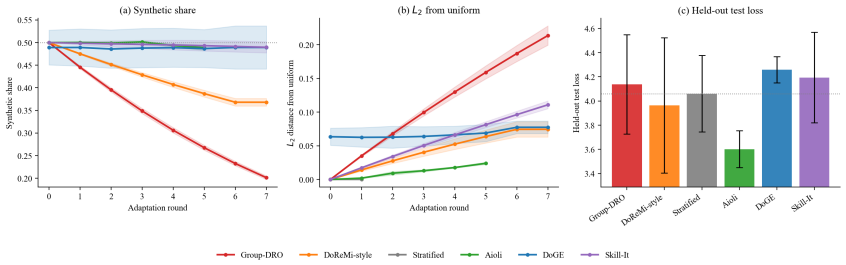


Figure 1: Central result (bands are $\pm 1\sigma$ over seeds 0, 1, 2). (a) Synthetic share ($\sum p$ over the 9 synthetic domains) vs. reweighting round, start $= 0.50$: loss-monotone methods move down, interaction and gradient estimators stay near uniform. (b) L_2 distance of the final mixture from uniform vs. round. (c) Held-out test loss by method (mean $\pm 1\sigma$ over seeds): Aioli has the lowest mean, but at $n = 3$ on online-trained models this is a consistency read, not a ranking.

carry to another (Kaplan et al., 2020). Reusing a finite pool under a compute ceiling is the data-constrained and compute-optimal scaling regime (Hoffmann et al., 2022; Muennighoff et al., 2023).

Evaluation and metrics. Alongside the two policy metrics we report held-out test loss over the 18 domains, broken out into real-only and synthetic-only sub-means. Training uses AdamW (lr 5×10^{-4}, linear warmup) with seeds 0, 1, 2. See Appendix F for the config block and artifact list.

3 Results and analysis

The Full-tier S2 runs used three seeds (0, 1, 2); headline numbers average across them, with each seed broken out in Appendix C.

Which methods depart from uniform at this scale. The policy split is clean and survives all three seeds (Figure 1a; per-seed in Appendix C). Methods keyed to loss drag the synthetic share down: Group-DRO reaches 0.20, the DoReMi-style variant 0.37. Aioli, Skill-It, and DoGE hardly leave their 0.50 start, sitting near 0.49. DoGE is the least stable, with its share wandering from 0.453 to 0.557 between seeds, a spread no other method shows. It traces to the self term in DoGE's score (Appendix A), which leaks a domain's own loss whenever it outweighs the cross terms.

The L_2-from-uniform read (Figure 1b; computed per seed then averaged, Appendix B) counts how much mass a method moves, direction aside: Group-DRO (0.214) > Skill-It

(0.111) > DoGE (0.077) ≈ DoReMi-style (0.074) > Aioli (0.024) > Stratified (0). Skill-It is the oddity. Its share hardly moves, yet its L_2 lands second, because it churns weight inside each provenance class rather than across the line (Appendix B).

Does real-versus-synthetic change the ranking, and why. Only the loss-monotone methods move much; the rest barely budge. Training pries open a loss gap between the two provenances, and the loss-movers chase it. Beforehand there's no gap to chase. At round 0 the provenances sit nearly tied (real 2.94, synthetic 2.90). One adaptation round later, real text is the high-loss side and templated synthetic the low-loss side, ending about one nat apart (final-round real 4.09, synthetic 3.06); we read the climb on real text as forgetting on the small fixed real shards (Figure 4). So it's not provenance the loss-movers chase. Group-DRO sheds low-loss real domains like Github and ArXiv and piles onto high-loss ones; what steers the weights is highest loss (Appendix B).

The held-out test loss, hedged. Treat the quality read (Figure 1c; real and synthetic breakdown in Figure 5; full means, sub-means, and per-seed values in Appendix C) cautiously. With $n = 3$ it tells us about average behavior, not an ordering, because the models were scored after online training rather than a clean static retrain. Aioli has the lowest aggregate mean (3.60 nats) and beats DoGE and Skill-It on every seed; we take this as consistent with the mechanism, not a clean quality verdict, since online dynamics and Aioli's probe-driven trajectory are confounded with mixture choice. Against Stratified, DoReMi, and Group-DRO it lands within seed scatter, and no aggressive loss-mover pulls ahead of the uniform control by more than that scatter. Whichever method reallocates least wins the mean; the heaviest reallocators never clear uniform, matching Section 4.

4 Mechanism in brief

What separates the families is how much the update's score leans on a domain's own loss. Group-DRO uses that loss straight. DoReMi instead measures excess loss against a reference, which dodges the pull toward irreducible noise. Aioli and DoGE ask, instead, how much training on a domain lowers loss elsewhere. Aioli's column sum stays purely cross-domain; DoGE's gradient inner product carries a self term that still tracks own loss. So DoGE escapes the loss-monotone pull only once cross-domain terms dominate, not by construction. This buys us an a-priori prediction (Appendix A): a loss-monotone method drops the synthetic share exactly when real is the higher-loss provenance. When real turns hard and synthetic easy, loss-driven scores cut synthetic share where the interaction methods sit still, with sign $\text{sign}(\bar{\ell}_{\text{real}} - \bar{\ell}_{\text{synth}})$, where $\bar{\ell}$ is the mean per-provenance validation loss. Derivation, loss-is-not-value decomposition, and out-of-sample corollary live in Appendix A.

5 Limitations and the compute floor

Two things deserve doubt: the protocol's signal and its budget. The quality read comes from online-trained models, not a static retrain, so it tangles the choice of mixture with online dynamics, plus probe overhead in Aioli's case; we hedge it rather than rank on it. For a verdict you'd retrain statically on each final mixture. Three seeds, one scenario, one model size (S2, 160M). The synthetic class is defined by instruction format rather than strictly by model generation (Section 2), so provenance and template are confounded; the one synthetic slot whose loss overlaps the real band, Self-Instruct (4.03; Figure 4), is the human-written Dolly-15k. Split and L_2 ordering hold tight apart from DoGE, which we judge by its mean and update rule, not any single seed. The two cheaper tiers (Table 1, Figure 2; Appendix D) locate a floor. The Intermediate tier (1.2×10^{16} FLOPs on pythia-70m) already reproduces the Full-tier split. At the Minimal tier (2.5×10^{14}, a minutes-long smoke test) it falls apart: across seeds Group-DRO points nowhere on net (std 0.35) and DoReMi-style actually climbs, ending at 0.75. We read this as seed-dominated, not proven noise; at $n = 3$ the two are hard to pull apart. The floor is the lowest budget where the loss-monotone methods agree in direction across seeds: Group-DRO's three Intermediate seeds land in $[0.234, 0.239]$, but at the Minimal tier they scatter to 0.996, 0.253, and 0.262. It sits between the two tiers. Both cheaper tiers run pythia-70m, so the floor is a budget effect at fixed model size, not a size

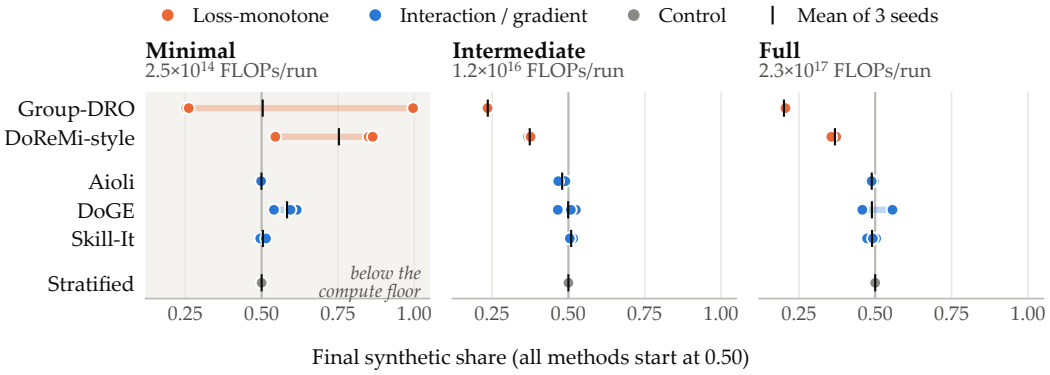


Figure 2: The compute floor: final synthetic share per method and tier (dots: seeds; bar: range; tick: mean). Below the floor the loss-monotone methods are seed-dominated (Group-DRO's seeds land at 0.253, 0.262, 0.996) or reversed (DoReMi-style); from the Intermediate tier on they sit tight below the 0.50 start, while the interaction and gradient methods stay near it.

effect; the Full tier (pythia-160m) only confirms it above, and an intermediate 160m point is untested. From roughly 10^{16} FLOPs the phenomenon reads clearly, three orders below a frontier run. Frontier compute isn't the requirement, just a modest floor. Whether the split survives past 160M we leave open.

6 Conclusion

Split a pool by provenance, pin the signal, read off the synthetic share: at 160M over three seeds, six data-mixing methods sort into two camps. Loss-monotone ones chase the hard real text downward; the rest sit near uniform, the steadiest (Aioli) posting the lowest mean held-out loss. Nobody reliably beats uniform; the protocol logs mixture choice, not quality, a reading already legible from about 10^{16} FLOPs (Section 5).

Using the probe. To vet a new reweighter before any large sweep, run it beside Stratified and Group-DRO on a provenance-split pool at the Intermediate budget or above, over three seeds: about 1.1×10^{17} FLOPs in all, less than one Full-tier run. Read each method's synthetic-share direction and its seed std. A share that falls with Group-DRO's, as tightly (std under 0.01), marks a loss-monotone policy; a share within scatter of 0.50 with a growing L_2, as for Skill-It, marks within-provenance reshuffling. Below the floor, where Group-DRO itself scatters, report the run as inconclusive, not as a null.

Acknowledgments

We thank the Aioli authors for releasing their reference implementation, which our port follows, and EleutherAI for the open Pythia checkpoints.

Ethics Statement

We continue training small public checkpoints on public HuggingFace corpora (Appendix F), used under their published terms for non-commercial research, and release code and aggregate statistics (losses, mixture weights, FLOPs), not raw text or model weights. Some synthetic corpora (e.g., Alpaca) are distilled from commercial models whose terms restrict downstream use; ShareGPT holds user-shared chatbot conversations that may contain personal information; the Pile contains web text of mixed copyright status, and our nine subsets exclude Books3. Our findings concern mixture choice only, not the safety of the resulting models.

References

Amro Abbas, Kushal Tirumala, Dániel Simig, Surya Ganguli, and Ari S. Morcos. SemDeDup: Data-efficient learning at web-scale through semantic deduplication, 2023. arXiv preprint arXiv:2303.09540.

Alon Albalak, Liangming Pan, Colin Raffel, and William Yang Wang. Efficient online data mixing for language model pre-training. *arXiv preprint arXiv:2312.02406*, 2023.

Alon Albalak, Yanai Elazar, Sang Michael Xie, Shayne Longpre, Nathan Lambert, Xinyi Wang, Niklas Muennighoff, Bairu Hou, Liangming Pan, Haewon Jeong, Colin Raffel, Shiyu Chang, Tatsunori Hashimoto, and William Yang Wang. A survey on data selection for language models. *Transactions on Machine Learning Research*, 2024.

Sanjeev Arora, Elad Hazan, and Satyen Kale. The multiplicative weights update method: a meta-algorithm and applications. *Theory of Computing*, 8(6):121–164, 2012. doi: 10.4086/toc.2012.v008a006.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning (ICML)*, pp. 41–48. ACM, 2009. doi: 10.1145/1553374.1553380.

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, et al. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pp. 2397–2430. PMLR, 2023.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 1877–1901, 2020. URL https://arxiv.org/abs/2005.14165.

Mayee F. Chen, Nicholas Roberts, Kush Bhatia, Jue Wang, Ce Zhang, Frederic Sala, and Christopher Ré. Skill-it! a data-driven skills framework for understanding and training language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023.

Mayee F. Chen, Michael Y. Hu, Nicholas Lourie, Kyunghyun Cho, and Christopher Ré. Aioli: A unified optimization framework for language model data mixing. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://openreview.net/forum?id=sZGZJhaNSe. arXiv:2411.05735.

Daixuan Cheng, Yuxian Gu, Shaohan Huang, Junyu Bi, Minlie Huang, and Furu Wei. Instruction pre-training: Language models are supervised multitask learners. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. URL https://arxiv.org/abs/2406.14491.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, et al. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25 (70):1–53, 2024. URL https://www.jmlr.org/papers/v25/23-0870.html.

Mike Conover, Matt Hayes, Ankit Mathur, Jianwei Xie, Jun Wan, Sam Shah, Ali Ghodsi, Patrick Wendell, Matei Zaharia, and Reynold Xin. Free Dolly: Introducing the world's first truly open instruction-tuned LLM. Databricks blog; dataset at https://huggingface.co/datasets/databricks/databricks-dolly-15k, 2023.

Shizhe Diao, Yu Yang, Yonggan Fu, Xin Dong, Dan Su, Markus Kliegl, Zijia Chen, Peter Belcak, Yoshi Suhara, Hongxu Yin, Mostofa Patwary, Yingyan Lin, Jan Kautz, and Pavlo Molchanov. Nemotron-CLIMB: Clustering-based iterative data mixture bootstrapping for language model pre-training. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025. URL https://openreview.net/forum?id=aBlqKPkc4a.

Ning Ding, Yulin Chen, Bokai Xu, Yujia Qin, Shengding Hu, Zhiyuan Liu, Maosong Sun, and Bowen Zhou. Enhancing chat language models by scaling high-quality instructional conversations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 3029–3051. Association for Computational Linguistics, 2023. URL https://aclanthology.org/2023.emnlp-main.183/.

Logan Engstrom, Axel Feldmann, and Aleksander Madry. DsDm: Model-aware dataset selection with datamodels. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 12491–12526. PMLR, 2024. URL https://proceedings.mlr.press/v235/engstrom24a.html.

Simin Fan, Matteo Pagliardini, and Martin Jaggi. DoGE: Domain reweighting with generalization estimation. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 12895–12915. PMLR, 2024. URL https://proceedings.mlr.press/v235/fan24e.html.

Yoav Freund and Robert E. Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1):119–139, 1997. doi: 10.1006/jcss.1997.1504.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, et al. The pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2021. URL https://arxiv.org/abs/2101.00027.

Ce Ge, Zhijian Ma, Daoyuan Chen, Yaliang Li, and Bolin Ding. BiMix: A bivariate data mixing law for language model pretraining, 2024. arXiv preprint arXiv:2405.14908.

Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Dhruv Pai, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David L. Donoho, and Sanmi Koyejo. Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. In *First Conference on Language Modeling (COLM)*, 2024. URL https://openreview.net/forum?id=5B2K4LRgmz.

Amirata Ghorbani and James Zou. Data shapley: Equitable valuation of data for machine learning. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pp. 2242–2251. PMLR, 2019. URL https://proceedings.mlr.press/v97/ghorbani19c.html.

Suriya Gunasekar, Yi Zhang, Jyoti Aneja, Caio César Teodoro Mendes, Allie Del Giorno, Sivakanth Gopi, Mojan Javaheripi, Piero Kauffmann, Gustavo de Rosa, Olli Saarikivi, et al. Textbooks are all you need. *arXiv preprint arXiv:2306.11644*, 2023.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2203.15556.

Yiding Jiang, Allan Zhou, Zhili Feng, Sadhika Malladi, and J. Zico Kolter. Adaptive data optimization: Dynamic sample selection with scaling laws. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://openreview.net/forum?id=aqok1UX7Z1.

Feiyang Kang, Newsha Ardalani, Michael Kuchnik, Youssef Emad, Mostafa Elhoushi, Shubhabrata Sengupta, Shang-Wen Li, Ramya Raghavendra, Ruoxi Jia, and Carole-Jean Wu. Demystifying synthetic data in LLM pre-training: A systematic study of scaling laws, benefits, and pitfalls. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 10739–10758. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.544. URL https://aclanthology.org/2025.emnlp-main.544/.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Jyrki Kivinen and Manfred K. Warmuth. Exponentiated gradient versus gradient descent for linear predictors. *Information and Computation*, 132(1):1–63, 1997. doi: 10.1006/inco.1996.2612.

Ariel N. Lee, Cole J. Hunter, and Nataniel Ruiz. Platypus: Quick, cheap, and powerful refinement of LLMs, 2023. arXiv preprint arXiv:2308.07317.

Jeffrey Li, Alex Fang, Georgios Smyrnis, Maor Ivgi, Matt Jordan, Samir Gadre, Hritik Bansal, Etash Guha, Sedrick Keh, Kushal Arora, et al. DataComp-LM: In search of the next generation of training sets for language models. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024.

Yuanzhi Li, Sébastien Bubeck, Ronen Eldan, Allie Del Giorno, Suriya Gunasekar, and Yin Tat Lee. Textbooks are all you need II: phi-1.5 technical report. *arXiv preprint arXiv:2309.05463*, 2023. URL https://arxiv.org/abs/2309.05463.

Qian Liu, Xiaosen Zheng, Niklas Muennighoff, Guangtao Zeng, Longxu Dou, Tianyu Pang, Jing Jiang, and Min Lin. RegMix: Data mixture as regression for language model pre-training. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2407.01492. Spotlight.

Shayne Longpre, Gregory Yauney, Emily Reif, Katherine Lee, Adam Roberts, Barret Zoph, Denny Zhou, Jason Wei, Kevin Robinson, David Mimno, and Daphne Ippolito. A pretrainer's guide to training data: Measuring the effects of data age, domain coverage, quality, and toxicity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, pp. 3245–3276. Association for Computational Linguistics, 2024. URL https://aclanthology.org/2024.naacl-long.179/.

Zheheng Luo, Xin Zhang, Xiao Liu, Haoling Li, Yeyun Gong, Qi Chen, and Peng Cheng. Velocitune: A velocity-based dynamic domain reweighting method for continual pre-training. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 16644–16656. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.813. URL https://aclanthology.org/2025.acl-long.813/. arXiv:2411.14318.

Pratyush Maini, Skyler Seto, Richard Bai, David Grangier, Yizhe Zhang, and Navdeep Jaitly. Rephrasing the web: A recipe for compute and data-efficient language modeling. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 14044–14072. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.757. URL https://aclanthology.org/2024.acl-long.757/.

Niklas Muennighoff, Alexander M. Rush, Boaz Barak, Teven Le Scao, Aleksandra Piktus, Nouamane Tazi, Sampo Pyysalo, Thomas Wolf, and Colin Raffel. Scaling data-constrained language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2305.16264.

Hongseok Namkoong and John C. Duchi. Stochastic gradient methods for distributionally robust optimization with f-divergences. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. URL https://proceedings.neurips.cc/paper/2016/hash/4588e674d3f0faf985047d4c3f13ed0d-Abstract.html.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL https://arxiv.org/abs/2203.02155.

Rui Pan, Dylan Zhang, Hanning Zhang, Xingyuan Pan, Minrui Xu, Jipeng Zhang, Renjie Pi, Xiaoyu Wang, and Tong Zhang. ScaleBiO: Scalable bilevel optimization for LLM data reweighting. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 31959–31982. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.1543. URL https://aclanthology.org/2025.acl-long.1543/.

Guilherme Penedo, Quentin Malartic, Daniel Hesslow, Ruxandra Cojocaru, Alessandro Cappelli, Hamza Alobeidli, Baptiste Pannier, Ebtesam Almazrouei, and Julien Launay. The RefinedWeb dataset for Falcon LLM: Outperforming curated corpora with web data, and web data only. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. URL https://openreview.net/forum?id=kM5eGcdCzq.

Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024. URL https://openreview.net/forum?id=n6SCkn2QaG.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI technical report*, 2019. URL https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL https://www.jmlr.org/papers/v21/20-074.html.

Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations (ICLR)*, 2020. URL https://arxiv.org/abs/1911.08731.

Victor Sanh, Albert Webson, Colin Raffel, Stephen H. Bach, Lintang Sutawika, et al. Multitask prompted training enables zero-shot task generalization. In *The Tenth International Conference on Learning Representations (ICLR)*, 2022. URL https://openreview.net/forum?id=9Vrb9D0WI4.

Zhiqiang Shen, Tianhua Tao, Liqun Ma, Willie Neiswanger, Zhengzhong Liu, Hongyi Wang, Bowen Tan, Joel Hestness, Natalia Vassilieva, Daria Soboleva, and Eric Xing. SlimPajama-DC: Understanding data combinations for LLM training, 2023. arXiv preprint arXiv:2309.10818.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget, 2023. arXiv preprint arXiv:2305.17493.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631(8022): 755–759, 2024. doi: 10.1038/s41586-024-07566-y.

Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos. Beyond neural scaling laws: Beating power law scaling via data pruning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL https://arxiv.org/abs/2206.14486.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following LLaMA model. https://github.com/tatsu-lab/stanford_alpaca, 2023.

Tristan Thrush, Christopher Potts, and Tatsunori Hashimoto. Improving pretraining data using perplexity correlations. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://openreview.net/forum?id=huuKoVQnB0.

Anvith Thudi, Evianne Rovers, Yangjun Ruan, Tristan Thrush, and Chris J. Maddison. MixMin: Finding data mixtures via convex minimization. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*. PMLR, 2025.

Kushal Tirumala, Daniel Simig, Armen Aghajanyan, and Ari S. Morcos. D4: Improving LLM pretraining via document de-duplication and diversification. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. URL https://openreview.net/forum?id=CG0L2PFrb1.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. LLaMA: Open and efficient foundation language models, 2023. arXiv preprint arXiv:2302.13971.

Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Atharva Naik, Arjun Ashok, et al. Super-naturalinstructions: Generalization via declarative instructions on 1600+ NLP tasks. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 5085–5109. Association for Computational Linguistics, 2022. URL https://aclanthology.org/2022.emnlp-main.340/.

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 13484–13508. Association for Computational Linguistics, 2023. URL https://aclanthology.org/2023.acl-long.754/.

Maurice Weber, Daniel Y. Fu, Quentin Anthony, Yonatan Oren, Shane Adams, Anton Alexandrov, Xiaozhong Lyu, Huu Nguyen, Xiaozhe Yao, Virginia Adams, Ben Athiwaratkun, Rahul Chalamala, Kezhen Chen, Max Ryabinin, Tri Dao, Percy Liang, Christopher Ré, Irina Rish, and Ce Zhang. RedPajama: an open dataset for training large language models. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024. URL https://arxiv.org/abs/2411.12372.

Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. Finetuned language models are zero-shot learners. In *The Tenth International Conference on Learning Representations (ICLR)*, 2022a. URL https://openreview.net/forum?id=gEZrGCozdqR.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent abilities of large language models. *Transactions on Machine Learning Research (TMLR)*, 2022b. URL https://openreview.net/forum?id=yzkSU5zdwD.

Alexander Wettig, Aatmik Gupta, Saumya Malik, and Danqi Chen. QuRating: Selecting high-quality data for training language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 52915–52971. PMLR, 2024. URL https://proceedings.mlr.press/v235/wettig24a.html.

Mengzhou Xia, Tianyu Gao, Zhiyuan Zeng, and Danqi Chen. Sheared LLaMA: Accelerating language model pre-training via structured pruning. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024a. URL https://openreview.net/forum?id=09iOdaeOzp.

Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. LESS: Selecting influential data for targeted instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 54104–54132. PMLR, 2024b. URL https://proceedings.mlr.press/v235/xia24c.html.

Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and Adams Wei Yu. DoReMi: Optimizing data mixtures speeds up language model pretraining. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023a. URL https://arxiv.org/abs/2305.10429.

Sang Michael Xie, Shibani Santurkar, Tengyu Ma, and Percy Liang. Data selection for language models via importance resampling. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023b. URL https://arxiv.org/abs/2302.03169.

Wanyun Xie, Francesco Tonin, and Volkan Cevher. Chameleon: A flexible data-mixing framework for language model pretraining and finetuning. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. PMLR, 2025. URL https://arxiv.org/abs/2505.24844.

Can Xu, Qingfeng Sun, Kai Zheng, Xiubo Geng, Pu Zhao, Jiazhan Feng, Chongyang Tao, Qingwei Lin, and Daxin Jiang. WizardLM: Empowering large pre-trained language models to follow complex instructions. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=CfXh93NDgH.

Zitong Yang, Neil Band, Shuangping Li, Emmanuel Candès, and Tatsunori Hashimoto. Synthetic continued pretraining. In *International Conference on Learning Representations (ICLR)*, 2025.

Jiasheng Ye, Peiju Liu, Tianxiang Sun, Jun Zhan, Yunhua Zhou, and Xipeng Qiu. Data mixing laws: Optimizing data mixtures by predicting language modeling performance. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2403.16952.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, Todor Mihaylov, Myle Ott, Sam Shleifer, Kurt Shuster, Daniel Simig, Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. OPT: Open pre-trained transformer language models, 2022. arXiv preprint arXiv:2205.01068.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. URL https://openreview.net/forum?id=uccHPGDlao.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. LIMA: Less is more for alignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2305.11206.

A Theoretical background

Setup. Let domains be indexed by $i = 1, \ldots, k$, with mixture weights $p \in \Delta^{k-1}$ (the probability simplex) and per-domain validation loss $\ell_i(\theta)$. Online reweighting methods update p across rounds and train on the mixture between updates. The methods we compare differ only in the score that drives a multiplicative-weights (exponentiated-gradient) update of the form

$$p_i^{(t+1)} \;\propto\; p_i^{(t)} \exp\big(\eta\, s_i^{(t)}\big), \tag{1}$$

where η is a step size and s_i is the method-specific score. The families differ in whether s_i is a function of ℓ_i alone.

Family 1: loss-monotone (Group-DRO, the DoReMi inner step). Here the score is monotone increasing in a domain's own (excess) loss:

$$\text{Group-DRO:} \quad s_i = \ell_i - \tfrac{1}{k}\textstyle\sum_j \ell_j, \tag{2}$$

$$\text{DoReMi:} \quad s_i = \big(\ell_i - \ell_i^{\text{ref}}\big) - \tfrac{1}{k}\textstyle\sum_j \big(\ell_j - \ell_j^{\text{ref}}\big), \tag{3}$$

where ℓ_i^{ref} is a reference-model loss. Equation (1) with the Group-DRO score is online mirror descent on the worst-case (distributionally robust) objective

$$\max_{p \in \Delta^{k-1}} \; \sum_{i=1}^{k} p_i\, \ell_i, \tag{4}$$

whose maximizer concentrates mass on $\arg\max_i \ell_i$ (Sagawa et al., 2020). Plain worst-case (raw-loss) optimization therefore climbs toward the highest-loss domains, which on a heterogeneous pool are the noisiest, highest-irreducible-loss ones; our Group-DRO baseline is exactly this raw-loss objective. DoReMi differs in an important way: it scores the *excess* loss over a reference, $\ell_i - \ell_i^{\text{ref}}$, precisely so that a domain that is hard for everyone (high ℓ_i but also high ℓ_i^{ref}) does not dominate; the excess-loss centering is what counters the raw-loss pull toward irreducible noise (Xie et al., 2023a). Our DoReMi-style proxy keeps the excess-loss form but, like Group-DRO, drops the joint min-max model step, so it sits between the two; this is why it moves the synthetic share down (to 0.37) less than raw Group-DRO (to 0.20). Its reference is a fixed pretrained base, provenance-neutral at the start, so subtracting it preserves the training-induced gap and the proxy stays in the loss-monotone family; a reference further trained on the pool would move off provenance-neutral and need not stay loss-monotone, so the grouping is contingent on the frozen reference.

Family 2: interaction / gradient (Aioli, DoGE). Here the signal is cross-domain, not ℓ_i. Aioli (Linear Mixing Optimization) models the loss dynamics as a linear mixing law,

$$\ell_i^{(t+1)} \;\approx\; \ell_i^{(t)} - \sum_j A_{ij}\, p_j, \qquad A_{ij} = \text{effect of training on } j \text{ on the loss of } i, \tag{5}$$

with A estimated by near-one-hot probe sweeps. The exploit step is exponentiated-gradient descent on the *column sums* of A:

$$p_j^{(t+1)} \;\propto\; p_j^{(t)} \exp\Big(\eta \textstyle\sum_i A_{ij}\Big). \tag{6}$$

It upweights domains j whose training most reduces the average loss (large column sum), not domains with high own loss. DoGE uses a first-order Taylor argument: a step on domain j moves θ by $-\eta\nabla\ell_j$, and the change in the universal-generalization target L_T (with gradient $g_T = \sum_i \nabla\ell_i$) is approximately $\eta\langle\nabla\ell_j, g_T\rangle$. So DoGE scores j by the gradient inner product

$$s_j = \Big\langle \nabla\ell_j,\; \textstyle\sum_i \nabla\ell_i \Big\rangle, \tag{7}$$

the gradient-space analogue of Aioli's column sums (Chen et al., 2025; Fan et al., 2024). Aioli's column sum is not a function of ℓ_j at all; DoGE's inner product expands into a self term $\|\nabla\ell_j\|^2$, correlated with domain j's own loss, plus cross terms $\sum_{i\neq j}\langle\nabla\ell_j, \nabla\ell_i\rangle$, so it is not a function of ℓ_j alone but is not strictly free of it either. DoGE's near-uniform behavior here is conditional on the cross terms dominating, which holds empirically on this pool but is not guaranteed by the score.

Family 3: hybrid (Skill-It). Skill-It learns a skills graph A (A_{ij} = effect of training skill i on eval skill j), then applies the multiplicative-weights update

$$w_i^{(t+1)} \propto w_i^{(t)} \exp\Big(\eta \textstyle\sum_j A_{ij}\, \ell_j\Big), \tag{8}$$

the current per-skill loss *weighted through the learned influence graph* (Chen et al., 2023). The loss enters, but only via which skills it helps, so the update is intermediate between the two families and, on this pool, reshuffles within rather than across the provenance boundary.

The a-priori prediction. A loss-monotone update (Eq. (1) with s_i monotone in ℓ_i) is forced toward the highest-loss provenance class; an interaction or gradient update (Eq. (6), Eq. (7)) is not, because its move depends on column sums of A, not on ℓ. Which provenance is higher-loss is an empirical input. On a provenance-split pool the two families therefore generically select different mixtures, and the sign of the synthetic-share movement equals

$$\operatorname{sign}\big(\bar{\ell}_{\text{real}} - \bar{\ell}_{\text{synth}}\big). \tag{9}$$

Measured on the online-trained model, $\bar{\ell}_{\text{real}} = 4.09 > \bar{\ell}_{\text{synth}} = 3.06$ (final round), so the loss-movers reduce the synthetic share while the interaction methods stay near uniform. An out-of-sample corollary: on a homogeneous all-real pool with no provenance loss gap, the families should not split. This is a cheap, falsifiable control we reserve for follow-up work; until it is run we describe the protocol as a candidate instrument, and a confirmed non-split would establish it as a reusable one.

Loss is not value (our synthesis). The DRO objective in Eq. (4) targets ℓ_i (the worst case), but the quantity governing generalization is the transfer $\sum_i A_{ij}$ that Aioli and DoGE estimate (Eq. (6), Eq. (7)); these coincide only if high loss implies high transfer, which is false in general. Decompose

$$\ell_i \;=\; H_i \;+\; \underbrace{(\ell_i - H_i)}_{\text{reducible gap}}, \tag{10}$$

where H_i is the irreducible entropy. A raw-loss method cannot separate the two terms, so it chases whichever provenance carries the higher loss whether or not that loss is reducible; here the real text becomes the high-loss provenance on the online-trained model and gets chased, even though high loss is not high value. DoReMi's excess-loss centering is the standard fix for exactly this failure of the raw-loss objective (Xie et al., 2023a), and Aioli's own finding is adjacent but distinct: that prior methods set their mixing-law parameters inaccurately, so loss-based mixing-law fits are unreliable predictors of good mixtures, and no prior method consistently beats stratified sampling (Chen et al., 2025). That a datum's loss differs from its value for generalization is also the premise of data-valuation and influence work (Ghorbani & Zou, 2019; Xia et al., 2024b), and selection signals keyed to the perplexity-to-downstream relationship rather than raw loss level reflect the same distinction (Thrush et al., 2025). We read "loss is not value" as our synthesis of these observations, not as a claim any of these papers makes verbatim. Our held-out test loss is consistent with it: the aggressive raw-loss movers do not beat the uniform baseline beyond seed noise, while the near-uniform interaction method (Aioli) has the lowest mean.

B Per-method weight evolution

L_2-from-uniform ordering and per-domain concentrations. The L_2 distance of the final mixture from uniform at the final round is Group-DRO 0.214 > Skill-It 0.111 > DoGE 0.077 ≈ DoReMi-style 0.074 > Aioli 0.024 > Stratified 0. Skill-It is the informative case: its synthetic share barely moves (0.49), yet its L_2 is second highest, because it reshuffles weight within each provenance class rather than across the real/synthetic boundary, so the synthetic-share number alone would mislabel it as inert. The per-domain weights are consistent (uniform = 0.0556): Group-DRO concentrates on high-loss real domains (Pile-CC 0.164, OpenWebText2 0.150) and starves easy synthetic ones (Code-Alpaca 0.004), while also

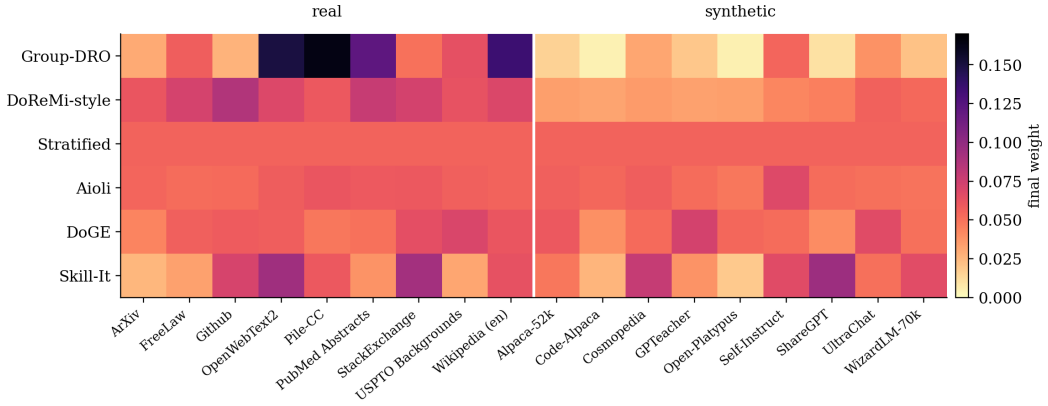


Figure 3: Final per-domain weights as a method-by-domain heatmap (mean over seeds; uniform = 0.0556), the real block left of the divider and the synthetic block right. Group-DRO is the most concentrated (dark cells on high-loss real domains, near-zero on easy synthetic ones); Aioli is the flattest non-trivial method.

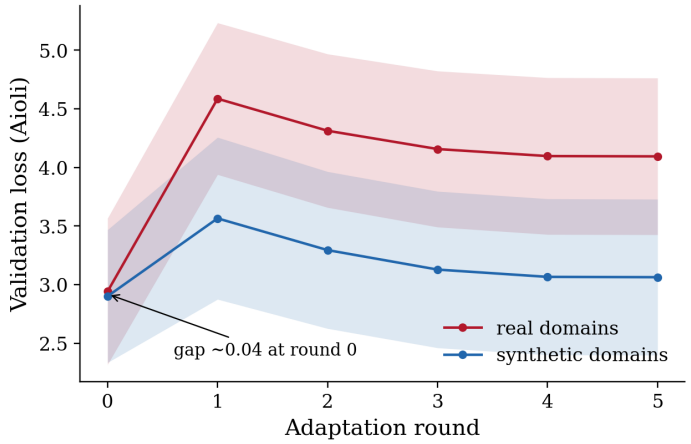


Figure 4: Per-provenance validation loss across rounds (Aioli run; line is the class mean, band the per-class spread). At round 0, the base checkpoint, real and synthetic are nearly tied (2.94 vs 2.90, gap 0.04); the gap opens inside the first adaptation round and settles near one nat (final-round real 4.09, synthetic 3.06), the real-domain rise consistent with forgetting on the small fixed real shards. At the final round the three lowest domains are synthetic (Code-Alpaca 2.00, Open-Platypus 2.23, ShareGPT 2.63) and the four highest real (Pile-CC 4.90, OpenWebText2 4.78, Wikipedia 4.62, PubMed 4.49), with Github (real, 2.90) and Self-Instruct (synthetic, 4.03) overlapping, so the split is systematic, not strict. This training-induced ordering is the empirical input that fixes the sign in Eq. (9).

trimming lower-loss real domains (Github 0.027, ArXiv 0.030), so the signal is highest-loss, not real as such; Aioli stays in a tight band (0.048 to 0.068).

The final per-domain weights confirm the reading. Figure 3 gives them as a method-by-domain heatmap and Figure 6 as grouped bars with per-seed error bands: Group-DRO concentrates aggressively on high-loss real domains and starves easy synthetic ones, Aioli stays in a tight near-uniform band, and DoGE is near-uniform but the noisiest. Figure 4 shows the Aioli per-domain validation-loss trajectories, where real domains sit high (mean 4.09) and synthetic domains sit low (mean 3.06), the empirical input that fixes the sign in Eq. (9).

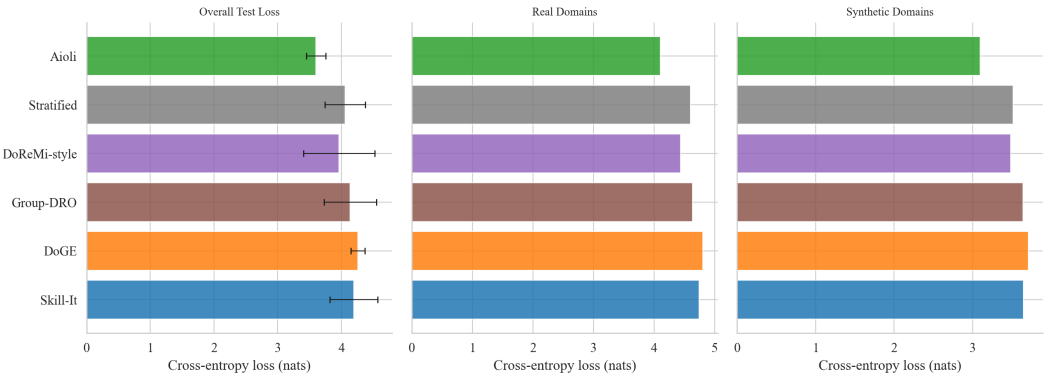


Figure 5: Held-out test loss disaggregated by provenance (Pythia-160m, Full tier, seeds 0, 1, 2). *Left*: aggregate mean cross-entropy loss (nats) over all 18 domains, with $\pm 1\sigma$ error bars across seeds. *Centre*: mean over the 9 real Pile subsets. *Right*: mean over the 9 synthetic instruction domains. Aioli achieves the lowest aggregate mean (3.60) and beats DoGE and Skill-It on all three seeds; it does not reliably beat Stratified, DoReMi-style, or Group-DRO at n=3, consistent with the average-case reading in Section 4. The consistent per-provenance gap, real domains scoring roughly one nat higher than synthetic for every method, independently confirms the training-induced loss ordering of Figure 4. The aggressive loss-movers (Group-DRO 4.14, DoReMi-style 3.96) do not improve over the uniform control (Stratified 4.06) in the aggregate mean, consistent with the mechanism in Section 4: chasing the high-loss real provenance reallocates mass but does not lower the balanced-pool test loss.

Table 2: Held-out test loss (lower is better), aggregate mean over 18 domains, mean $\pm$ std over 3 seeds, then real-only and synthetic-only sub-means. Read as an average-case signal at $n = 3$, not a ranking.

Method	test loss (mean $\pm$ std)	real	synth
Aioli	3.60 ± 0.15	4.10	3.10
DoReMi-style	3.96 ± 0.56	4.44	3.49
Stratified	4.06 ± 0.32	4.60	3.52
Group-DRO	4.14 ± 0.41	4.63	3.64
Skill-It	4.19 ± 0.37	4.74	3.65
DoGE	4.26 ± 0.11	4.81	3.71

C Complete per-seed tables

Table 2 gives the aggregate held-out test loss with real/synthetic sub-means, Table 3 the per-seed synthetic-share endpoints, and Table 4 the per-seed aggregate test loss.

The synthetic-share split is tight across seeds for every method except DoGE, whose share swings from 0.453 to 0.557. The test-loss read is the fragile one. Aioli has the lowest mean and beats DoGE and Skill-It on all three seeds, so the claim "Aioli has the best mean test loss and reliably beats DoGE and Skill-It" is defensible. But Aioli beats Stratified on only two of three seeds (Stratified is lower on seed 2, 3.62 vs 3.67) and is mixed against DoReMi and Group-DRO, so it does *not* reliably beat Stratified, DoReMi, or Group-DRO; we do not claim it is the only method to beat the uniform control or trains a definitively better model. The aggressive loss-movers are narrower still: neither beats uniform by more than noise (Group-DRO is worse on the mean, 4.14 vs 4.06; DoReMi-style lower by only 0.10 with std 0.56, inside the scatter). The consistent real-versus-synthetic gap across every method (real about 4.1 to 4.8, synthetic about 3.1 to 3.7) independently confirms that the templated synthetic text is the easier (lower-loss) provenance on the online-trained models, consistent with the training-induced gap of Figure 4.

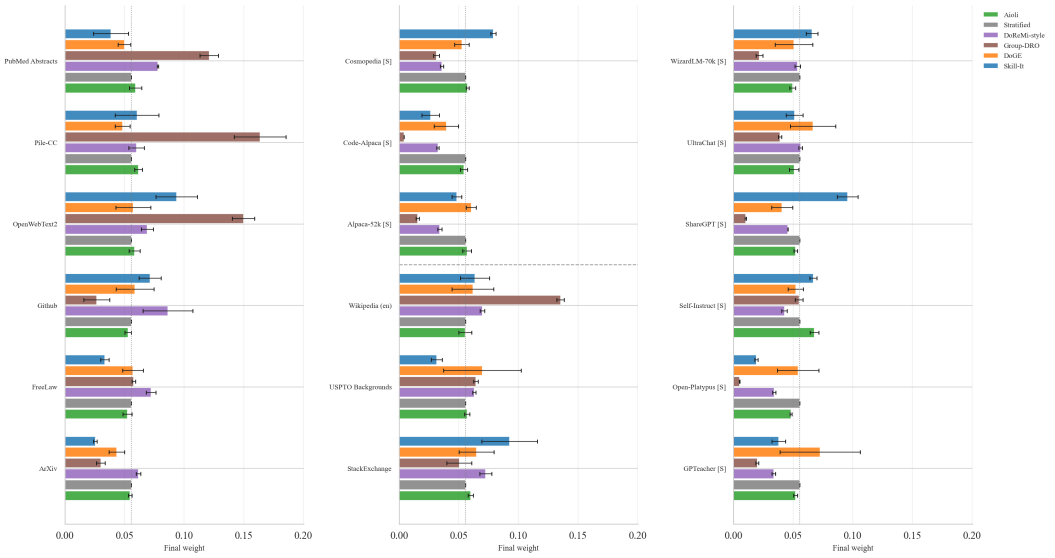


Figure 6: Final per-domain mixture weights as a grouped horizontal bar chart (mean $\pm 1\sigma$ over seeds 0, 1, 2; uniform $= 0.0\overline{5}$), complementing the heatmap of Figure 3 with per-method error bars. Domains are sorted alphabetically within each provenance class. Group-DRO concentrates on the highest-loss real domains (Pile-CC 0.164, OpenWebText2 0.150) and drives several synthetic domains near zero (Code-Alpaca 0.004), while also trimming the low-loss real domains (Github 0.027, ArXiv 0.030): the discriminating signal is per-domain loss magnitude, not provenance category per se. DoReMi-style shows the same directional pattern at attenuated magnitude. Aioli stays in a narrow band around uniform (approximately 0.04 to 0.07) with small cross-seed variance. Skill-It is the most informative case: individual domains within each provenance class deviate substantially from uniform, ShareGPT elevated in the synthetic block and OpenWebText2 and StackExchange elevated in the real block, yet the total mass on each class remains near 0.50, the within-class reshuffling behavior visible at the domain level (see also Figure 7).

Table 3: Per-seed synthetic-share endpoints (start = 0.50).

Method	seed 0	seed 1	seed 2	mean
Group-DRO	0.197	0.201	0.206	0.20
DoReMi-style	0.376	0.372	0.356	0.37
Stratified	0.500	0.500	0.500	0.50
Aioli	0.482	0.496	0.488	0.49
Skill-It	0.503	0.473	0.492	0.49
DoGE	0.557	0.453	0.458	0.49

D Cross-tier results: the compute floor

We run all three tiers of Table 1 over the same three seeds. The two pythia-70m tiers are not throwaway smoke tests; they locate where on the compute axis the loss-monotone split becomes legible. Figure 8 plots the synthetic-share trajectories at each budget, Figure 2 (Section 5) shows every seed's endpoint at each budget, and three tables give the numbers: final synthetic share (Table 5), held-out test loss (Table 6), and per-seed share at the two cheaper tiers (Table 7).

The directional pattern holds above a floor and disappears below it. At the Full (2.3×10^{17} FLOPs) and Intermediate (1.2×10^{16}) the loss-monotone family pulls the synthetic share down (Group-DRO to 0.20 and 0.24, DoReMi-style to 0.37 and 0.37) and the interaction and gradient methods stay near uniform. At the Minimal tier (2.5×10^{14}, a minutes-long run

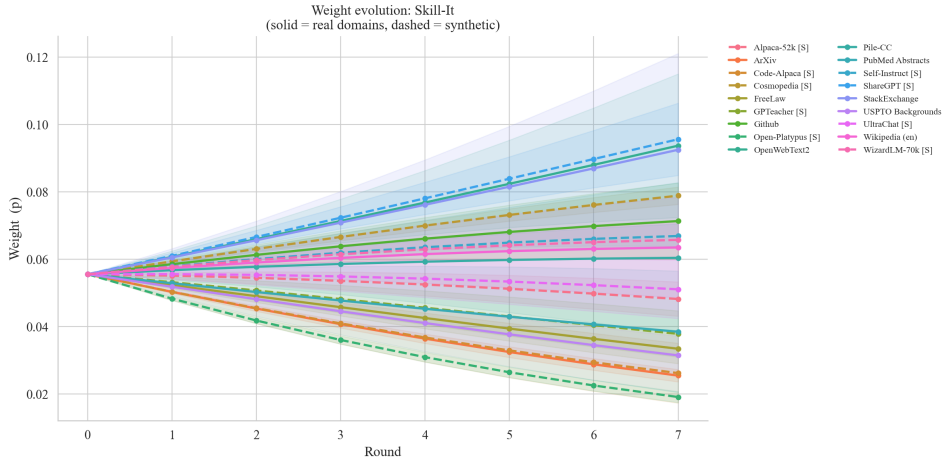


Figure 7: Skill-It per-domain weight trajectories across reweighting rounds (solid lines = real domains; dashed lines = synthetic domains; shaded bands = $\pm 1\sigma$ over seeds 0, 1, 2). Skill-It shows bidirectional, within-class movement: some synthetic domains gain weight (Cosmopedia, Self-Instruct) while others fall (Code-Alpaca, GPTeacher), and a symmetric pattern holds within the real class. The total synthetic share stays near 0.49 throughout even as the L_2 distance from uniform reaches 0.111 (second only to Group-DRO), making Skill-It the method that moves most without crossing the provenance boundary. This is the empirical signature of the graph-weighted update (Appendix A, Family 3): the learned influence graph A discriminates among domains within a provenance class more sharply than across the provenance boundary at this scale. The wider inter-seed bands on certain domains (StackExchange, OpenWebText2) indicate that specific within-class rankings are less seed-stable than the family-level synthetic-share result, consistent with the observations in Section 4.

Table 4: Per-seed aggregate held-out test loss (lower is better).

Method	seed 0	seed 1	seed 2	mean $\pm$ std
Aioli	3.39	3.74	3.67	3.60 ± 0.15
DoReMi-style	3.64	3.50	4.75	3.96 ± 0.56
Stratified	4.24	4.33	3.62	4.06 ± 0.32
Group-DRO	4.55	4.28	3.58	4.14 ± 0.41
DoGE	4.41	4.19	4.17	4.26 ± 0.11
Skill-It	3.68	4.36	4.55	4.19 ± 0.37

on pythia-70m) the split is absent. Group-DRO ends at 0.50 with a seed std of 0.35, and the per-seed values explain why: one seed drives the synthetic share to 0.996 while the other two send it down (Table 7). DoReMi-style runs the wrong way, ending at 0.75; its reversal is consistent across all three seeds (0.545, 0.850, 0.864), so below the floor it does not simply turn noisy the way Group-DRO does but enters a different regime, plausibly from extreme undertraining and fewer adaptation rounds. The seed std of the final share collapses with compute (Figure 2; Table 5), Group-DRO from 0.35 at the Minimal tier to under 0.01 by the Intermediate tier, so beneath the floor the loss-monotone direction is unreliable, whether through incoherent scatter (Group-DRO) or coherent reversal (DoReMi-style). A minimum budget is necessary to read the split, but a small one, about 10^{16} FLOPs, three orders below a frontier run.

The held-out test loss needs the budget too (Table 6). At the Minimal tier the 70m models are badly undertrained, losses of 7 to 10 nats with seed std up to 1.9, so no quality signal survives. At the Intermediate tier the six methods are statistically indistinguishable, all between 3.70 and 3.79; the Aioli-lowest-mean result (3.60 against a next-best 3.96) is a

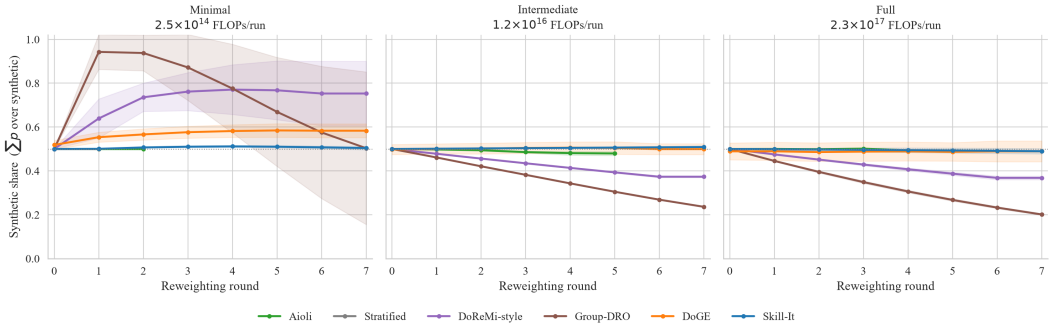


Figure 8: Synthetic-share trajectories at the three compute tiers (all methods, $\pm 1\sigma$ over seeds 0, 1, 2; start 0.50). At the Minimal tier the loss-monotone methods do not separate cleanly: Group-DRO spikes toward the synthetic class then collapses with a wide seed band, and DoReMi-style climbs to 0.75. From the Intermediate tier on, the split is the clean Full-tier pattern, loss-monotone methods down and the rest near uniform.

Table 5: Final synthetic share by tier (mean $\pm$ std over seeds 0, 1, 2; start 0.50). The loss-monotone split (Group-DRO, DoReMi-style move down) is present at the Intermediate and Full tiers but absent at the Minimal tier, where Group-DRO has no net direction (std 0.35) and DoReMi-style reverses. Interaction and gradient methods stay near 0.50 at every tier.

Method	Minimal (2.5×10^{14})	Intermediate (1.2×10^{16})	Full (2.3×10^{17})
Group-DRO	0.50 ± 0.35	0.24 ± 0.00	0.20 ± 0.00
DoReMi-style	0.75 ± 0.15	0.37 ± 0.00	0.37 ± 0.01
Aioli	0.50 ± 0.00	0.48 ± 0.01	0.49 ± 0.01
DoGE	0.58 ± 0.03	0.50 ± 0.03	0.49 ± 0.05
Skill-It	0.50 ± 0.01	0.51 ± 0.01	0.49 ± 0.01
Stratified	0.50 ± 0.00	0.50 ± 0.00	0.50 ± 0.00

Full-tier effect. The quality read, like the directional split, sharpens only with compute, which is why we hedge it even at the Full tier.

E Extended related work

This appendix places our study against the wider literature it draws on. The organizing thread is the paper's thesis: a domain's loss is not its value for generalization, the methods we compare split into a forced loss-monotone family and a free interaction/gradient family, and our protocol is a behavioral diagnostic on a fixed provenance-split pool rather than a quality ranking. We group the related work by theme and state for each entry the specific reason it bears on that thesis.

E.1 Data mixing and domain reweighting

Mixing methods are normally validated on homogeneous, all-real pools. DoReMi (Xie et al., 2023a) trains a reference and runs Group-DRO (Sagawa et al., 2020) on excess loss; RegMix (Liu et al., 2025) fits a regression over many small proxy runs; Aioli (Chen et al., 2025) unifies these as a mixing-law optimization and reports that prior methods set their mixing-law parameters inaccurately, so loss-based fits are unreliable predictors of good mixtures and no prior method consistently beats stratified sampling. DoGE (Fan et al., 2024) reweights by gradient alignment, Skill-It (Chen et al., 2023) by a learned skills graph, and Adaptive Data Optimization (Jiang et al., 2025) online by a scaling-law learning-potential signal rather than own loss; we implement DoGE and Skill-It from their definitions. The mixing-laws line fits a predictor and optimizes it: Data Mixing Laws (Ye et al., 2025) and BiMix (Ge et al.,

Table 6: Held-out test loss by tier (aggregate over 18 domains, mean $\pm$ std over seeds; lower is better). Minimal-tier 70m models are badly undertrained, so no quality signal survives; at the Intermediate tier the methods are within scatter (3.70 to 3.79); the Aioli-lowest-mean result is a Full-tier effect.

Method	Minimal	Intermediate	Full
Aioli	7.37 ± 0.77	3.70 ± 0.00	3.60 ± 0.15
DoReMi-style	9.54 ± 1.88	3.79 ± 0.04	3.96 ± 0.56
Stratified	7.33 ± 0.64	3.72 ± 0.02	4.06 ± 0.32
Group-DRO	8.57 ± 1.35	3.79 ± 0.01	4.14 ± 0.41
DoGE	7.50 ± 1.01	3.74 ± 0.02	4.26 ± 0.11
Skill-It	7.68 ± 0.18	3.70 ± 0.01	4.19 ± 0.37

Table 7: Per-seed final synthetic share at the two cheaper tiers (start 0.50). At the Minimal tier the loss-monotone methods are seed-dominated: Group-DRO sends one seed to 0.996 (toward synthetic) and two down, and DoReMi-style runs up on two of three. By the Intermediate tier the same methods are tight and pointed down.

	Minimal			Intermediate		
Method	seed 0	seed 1	seed 2	seed 0	seed 1	seed 2
Group-DRO	0.996	0.253	0.262	0.234	0.239	0.235
DoReMi-style	0.545	0.850	0.864	0.376	0.368	0.376
DoGE	0.540	0.615	0.594	0.524	0.465	0.508
Aioli	0.500	0.500	0.498	0.482	0.490	0.467
Skill-It	0.502	0.495	0.514	0.507	0.516	0.505
Stratified	0.500	0.500	0.500	0.500	0.500	0.500

2024) fit loss as a function of proportion (and, for BiMix, training steps), and MixMin (Thudi et al., 2025) finds mixtures by convex minimization. The salient contrast for a small-scale venue is compute cost on the axis we emphasize: RegMix fits a regression over many small proxy runs to predict the best mixture, MixMin solves a convex minimization, and a faithful DoReMi trains a reference, whereas our protocol reads policy behavior directly off one fixed pool with no proxy sweep or reference model.

Several recent methods sharpen what counts as the reweighting signal, which is exactly the axis our forced-versus-free split formalizes. Pan et al. (2025) casts data reweighting as a bilevel-optimization problem and shows that the online loss-based methods we study are cheap first-order approximations to that bilevel objective, which is why their behavior is governed by the per-domain loss signal we critique. Sheared-LLaMA (Xia et al., 2024a) pairs structured pruning with a "dynamic batch loading" rule that raises a domain's sampling rate in proportion to its per-domain loss gap against a reference, the loss-as-signal heuristic our diagnostic is built to expose: such a rule chases whichever provenance carries the higher loss. CLIMB (Diao et al., 2025) discovers mixtures by clustering documents in embedding space and searching over the induced cluster proportions, in contrast to our use of explicit provenance labels that keep the real/synthetic boundary visible to the policy. Chameleon (Xie et al., 2025) reweights from an embedding-space influence signal that is neither a loss nor a gradient inner product, which broadens the "what signal drives the update" axis beyond the loss-monotone and gradient families we group methods into and shows the design space is not exhausted by the two families on this pool.

E.2 Data selection and curation

Mixing is one half of a larger data-curation problem; the other is selection. Data selection picks which examples to keep, by importance resampling against a target distribution (Xie et al., 2023b) or by large-scale filtering experiments (Li et al., 2024), whereas mixing sets the proportions of fixed sources; the survey by Albalak et al. (2024) maps both. Our diagnostic also sits on the online side of an online-versus-offline divide: offline selection commits to a

corpus before training, while online data mixing adjusts proportions during training from a running signal (Albalak et al., 2023; Luo et al., 2025), which is the regime every method we compare operates in.

Corpus construction is itself a selection decision, and the curation literature repeatedly finds that what you keep, not raw loss, governs value. The C4 corpus (Raffel et al., 2020) fixes a filtering pipeline whose choices define the effective training distribution before any reweighting runs. RefinedWeb (Penedo et al., 2023) shows that carefully filtered web data can match or beat hand-curated corpora, and FineWeb (Penedo et al., 2024) reports that learned quality classifiers select better data than raw language-model loss, direct support for reading loss and value as distinct quantities. Upstream of any proportions, SemDeDup (Abbas et al., 2023) reshapes the effective distribution by removing semantic duplicates, and D4 (Tirumala et al., 2023) selects by embedding-based diversification, both showing that the composition a mixing policy then reweights is already a design choice. QuRating (Wettig et al., 2024) selects by learned human-style quality criteria rather than perplexity, and DsDm (Engstrom et al., 2024) makes selection model-aware and target-aware, finding that proximity to nominally "high-quality" sources can hurt, a selection-side echo of our point that low loss is not high value. Sorscher et al. (2022) argues that which examples you keep can matter more than how many, composition over raw scale, the same lever our methods automate on the proportions side. Closest to us methodologically, Longpre et al. (2024) runs a controlled diagnostic study of how pretraining composition affects downstream behavior; our work is the mixing-policy analogue, holding the pool fixed and reading off which policy keeps versus discards a provenance class.

E.3 Open multi-domain corpora

Our pool is built from provenance-tagged public sources. The Pile (Gao et al., 2021) supplies the nine real domains. RedPajama (Weber et al., 2024) is an open, provenance-tagged multi-domain corpus of the same kind, and the SlimPajama-DC study (Shen et al., 2023) shows that domain composition and deduplication materially change a 1.3B-parameter pretraining run, which is precisely the lever the mixing methods we compare adjust automatically; that the lever moves outcomes at the 1.3B scale is one reason a policy's silent reweighting at our scale is worth measuring.

E.4 Synthetic and instruction corpora as a provenance

The nine synthetic domains are instruction-formatted, assistant-style data, mostly model-generated, rather than raw web text, which anchors the real-versus-synthetic split that our diagnostic keeps visible. The slot we label Self-Instruct, after the bootstrapped instruction-generation recipe of Wang et al. (2023), is in fact filled by Dolly-15k (Conover et al., 2023), which human annotators wrote; Alpaca-52k (Taori et al., 2023) and the Evol-Instruct WizardLM-70k set (Xu et al., 2024) are model-distilled instruction corpora; UltraChat (Ding et al., 2023) is a large synthetic multi-turn dialogue corpus; Open-Platypus (Lee et al., 2023) is a curated instruction mixture drawn largely from human-authored problem sets; and the ShareGPT/Vicuna-style conversational data, human prompts with model responses, is of the kind benchmarked by MT-Bench (Zheng et al., 2023). Because every one of these is wrapped in the same instruction template and most are model-generated, they form a provenance class whose loss behaves differently from raw web text, which is what makes the provenance boundary a meaningful axis for the policy to act on.

E.5 Instruction-tuning background

Why instruction data is a distinct provenance, and not merely more text, is the subject of the instruction-tuning literature. FLAN (Wei et al., 2022a) shows that instruction tuning reshapes model behavior beyond what raw next-token loss predicts; T0 (Sanh et al., 2022) shows that the composition of a multitask prompted-training mixture governs downstream behavior; Super-NaturalInstructions (Wang et al., 2022) is a canonical large declarative-instruction dataset of the kind our synthetic slots emulate; InstructGPT (Ouyang et al., 2022) shows that the instruction and RLHF pipeline makes instruction corpora behaviorally distinct from

their pretraining text; and the scaled FLAN study (Chung et al., 2024) shows that scaling and composition of the instruction mixture, not just its volume, drive generalization. Together these motivate treating instruction data as its own provenance class rather than folding it into "more web text," so that a policy that drives the instruction share up or down is making a behaviorally loaded choice.

E.6 Synthetic-data effectiveness versus degradation

On the data side, the question of whether real and synthetic text should be combined at all is studied separately from how mixing algorithms behave on such a pool. Work on model collapse warns that training on recursively generated data degrades models (Shumailov et al., 2024), a finding first circulated in the originating preprint (Shumailov et al., 2023) before its journal version, but that accumulating real alongside synthetic data can break the curse of recursion (Gerstgrasser et al., 2024), which is one motivation for keeping a real/synthetic split visible to the mixing policy. On the other side, synthetic data can be strikingly effective: high-quality textbook-style synthetic corpora train small models well (Gunasekar et al., 2023), the phi-1.5 result extends this to general reasoning with synthetic textbook data substituting for web text at small scale (Li et al., 2023), mixing real text with synthetic rephrases is compute-efficient with gains attributable to style and format rather than lower loss (Maini et al., 2024), and synthetic continued pretraining extracts more from a fixed corpus (Yang et al., 2025). The instruction-tuning side makes the loss-versus-value gap explicit: a small high-quality instruction set can dominate far larger volumes (Zhou et al., 2023), decoupling a datum's value from its bulk and from any loss level. At larger scale the best synthetic fraction is bounded, converging near thirty percent with too much synthetic data hurting (Kang et al., 2025), independent corroboration that a policy silently driving the synthetic share to an extreme is not free. Instruction Pre-Training (Cheng et al., 2024) makes a concrete design choice in this space, converting a fraction of the corpus (about one fifth of documents, roughly ten percent of tokens) into instruction-formatted data and leaving the rest raw; we note the choice without importing any claim about an optimal fraction, since that paper does not ablate it. Our contribution sits orthogonal to all of these: we hold the pool fixed and ask which mixing policy keeps the synthetic share and which discards it.

E.7 Models and scaling background

The mixing methods we study refine a single-mixture pretraining recipe whose form was set by the foundational models. GPT-3 (Brown et al., 2020) established the few-shot model trained on one fixed mixture that reweighting methods later refine; LLaMA (Touvron et al., 2023) showed that careful composition of public data lets relatively small models compete, the regime our small-scale probe lives in; OPT (Zhang et al., 2022) released an open multi-size suite whose reproducibility goal parallels the Pythia (Biderman et al., 2023) checkpoints we use; and GPT-2 (Radford et al., 2019) framed unsupervised multitask learning from pretraining alone, the contrast against the supervised instruction data that forms our synthetic provenance. The scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) and the data-constrained regime (Muennighoff et al., 2023) fix the FLOP and token budgets our accounting reports against. The emergent-abilities observation (Wei et al., 2022b), that capability is not smoothly predictable from loss or scale, is both a direct statement of our "loss is not value" thesis and a scale caveat on our findings: a probe at our budget reads a behavioral direction, not the capability outcomes that may appear only at much larger scale.

E.8 Optimization-theory lineage of the shared update

Every method we compare drives the same multiplicative-weights, exponentiated-gradient update on the simplex (Eq. (1)); they differ only in the per-domain signal. Exponentiated gradient (Kivinen & Warmuth, 1997) is the canonical simplex-reweighting primitive that update instantiates, and the multiplicative-weights meta-algorithm (Arora et al., 2012) is the backbone the methods share, which is why isolating the score s_i is the right level at which to separate them into families. The historical root is the Hedge algorithm behind AdaBoost (Freund & Schapire, 1997), and the worst-case loss step in Group-DRO and DoReMi

Table 8: Domains (scenario S2). Synthetic examples are wrapped in a fixed instruction template, real text untemplated. The class is defined by instruction format, not purely by model generation: the slot labeled Self-Instruct (the label used in all figures) holds the human-written Dolly-15k, ShareGPT pairs human prompts with model responses, and Open-Platypus draws largely on human-authored problems.

Provenance	Domains
Real (9, Pile subsets)	Wikipedia (en), Pile-CC, OpenWebText2, Github, PubMed Abstracts, StackExchange, FreeLaw, USPTO Backgrounds, ArXiv
Synthetic (9, instruction)	UltraChat, Self-Instruct, Cosmopedia, Code-Alpaca, GPTeacher, WizardLM-70k, Alpaca-52k, ShareGPT, Open-Platypus

is grounded as a principled minimax procedure by the stochastic-gradient algorithm for f-divergence distributionally robust optimization of Namkoong & Duchi (2016). Curriculum learning (Bengio et al., 2009), ordering data by a difficulty-or-relevance signal, is the conceptual ancestor of Skill-It and of dynamic reweighting generally; framing the methods as one shared update with different scores is what lets our forced-versus-free split be a statement about the signal rather than about the optimizer.

F Reproducibility and configs

Setup. Tokenizer EleutherAI/pythia-160m (GPT-NeoX BPE, pad = eos). Models are HuggingFace Pythia pretrained checkpoints (not from scratch): the Full tier uses pythia-160m; the Intermediate and Minimal tiers use pythia-70m (Biderman et al., 2023). Optimizer AdamW, lr 5×10^{-4}, linear warmup, grad-clip 1.0, weight decay 0. Mixing is sampling: each train step draws a batch by multinomial(batch, p) over domains; per-domain shards are fixed-length token blocks. The per-domain validation loss is the signal every online method consumes. Seeds are 0, 1, 2; numbers are per-seed first, then aggregated.

Data sources. All corpora stream from the HuggingFace Hub. Real: the nine Pile subsets of `ola13/small-the_pile`. Synthetic: `HuggingFaceH4/ultrachat_200k` (UltraChat), `databricks/databricks-dolly-15k` (the slot labeled Self-Instruct), `HuggingFaceTB/cosmopedia`, `sahil2801/CodeAlpaca-20k`, `teknium/GPTeacher-General-Instruct`, `WizardLMTeam/WizardLM_evol_instruct_70k`, `tatsu-lab/alpaca`, `RyokoAI/ShareGPT52K`, and `garage-bAInd/Open-Platypus`.

Tiered configs. In the runner, `--config` selects the Minimal, Intermediate, and Full tiers as `SUPER_LIGHT`, `LIGHT`, and `HEAVY`. Minimal: pythia-70m, ctx 2048, batch 2, max_steps 144, update_steps 22, lp_rounds 3, lp_steps 1, 40k/8k/8k tok/domain (tr/val/te), no EMA. Intermediate: pythia-70m, ctx 1024, batch 4, max_steps 7200, update_steps 1100, lp_rounds 6, lp_steps 5, 200k/20k/20k, no EMA. Full (reported): pythia-160m, ctx 2048, batch 8, max_steps 14400, update_steps 2200, lp_rounds 6, lp_steps 10, 1M/100k/100k, EMA 0.1.

Method faithfulness (honest labels). Aioli (faithful port of HazyResearch/aioli; A orientation, sign, and EGD axis verified; the Full tier uses a stateful EMA path that deviates from the reference stateless reset). Stratified (trivially faithful control). DoReMi-style (approximate: val-shard loss, mean-centering, omits the $\max(0, \cdot)$ clip and $c = 10^{-3}$ smoothing; fixed pretrained reference, not further trained). Group-DRO (approximate: keeps the high-loss-upweight update, omits the joint min-max model step). DoGE (faithful to Fan et al. (2024); deviations: online single-model rather than proxy-to-base, per-round estimation, mean-abs score scaling). Skill-It (faithful to Chen et al. (2023), continual-pretraining regime; deviations: uniform init, mean-abs scaling). The older val-loss-improvement heuristics (Gen-Delta, Improvement-Window) survive only behind a flag as reference points and are never labeled DoGE/Skill-It.

FLOP surcharges and accounting caveats. The Full-tier base per-run budget is 2.30×10^{17} FLOPs ($6ND$, $N = 162{,}322{,}944$, $D = 2.36 \times 10^8$, including Aioli probe steps). Two methods carry a one-time surcharge over this base: DoGE adds 432 gradient-alignment micro-batches (+3%, total 2.37×10^{17}), and Skill-It adds one-time graph learning (18 single-skill proxies, 200 steps each; +25%, total 2.87×10^{17}). The DoReMi-style reference is a fixed pretrained checkpoint that is not further trained, so it adds no training FLOPs; a faithful DoReMi reference would add a second forward/backward stream, so our per-run number understates fully faithful DoReMi. Aioli probes by sweeping near-one-hot mixtures ($18 \cdot 6 \cdot 10 = 1080$ probe steps/round, inside max_steps), so at equal total FLOPs Aioli and the baselines do not perform equal exploit work; the figure is an equal-total-compute comparison, not an equal-exploit one. The headline S2 study is 6 methods $\times$ 3 seeds, about 18 runs, about 4.3×10^{18} FLOPs end to end (including the DoGE and Skill-It surcharges), each run roughly 435$\times$ under a 10^{20}-per-run ceiling. The diagnostic reads a direction and rough magnitude, stable to the exploit-step count, whereas any single absolute weight is not.

Extended limitations. On the EMA path, Aioli's reported run uses a stateful EMA that deviates from the reference stateless reset, so its exact endpoint is implementation-specific though its near-uniform position is not.

Released artifacts. We release the synthetic-share trajectories, per-domain weight CSVs, the L_2-from-uniform curves, the held-out test-loss CSV, the estimated A-matrices and p-history, the FLOP table, the tiered configs, and all comparison figures, together with the single self-contained runner (`run_experiment.py`) that regenerates them, so the family split can be reproduced and extended at the listed cost. Code and artifacts are available at `github.com/mixing-floor/compute-floor`; the runner exposes the three tiered configs and the held-out and static-retrain evaluation paths behind flags.